Breeding phenology, reproductive success and survival of Common Swift (*Apus apus*) from a Northern Italian colony

Milo Manica[1-2], Daniela Casola[2], Stocchetti Alessandra[2], Manuela Bazzarelli[3], Davide Biganzoli[4], Michelangelo Morganti[5-6]

1 LIPU Birdlife Italia, via Udine 3/A – 43122 Parma, 2 Gruppo Insubrico di Ornitologia, via Manzoni 21 – 21050 Clivio (VA), 3 via Puccini 8 – 21023 Besozzo (VA), 4 Comune di Jerago con Orago, via indipendenza 17, 5 CNR-IRSA Via del Mulino 19, 20861 Brugherio (MB), 6 GreenRes Sezione di Zoologia del MUSE, Corso del Lavoro e della Scienza, 3 – 38122, Trento (TN)

Introduction

Materials and methods

Results

Conclusion

Introduction

The reproductive biology of the Common Swift *Apus apus* has been widely studied, however to have a strong conservation program for the species we must increase the sample of colonies studied at a spatial level in the long term. This would enable us to understand phenomena that occur on a large scale as a consequence of climate change.

Since 2017, some associations together with the Municipality of Jerago con Orago (VA), Italy, have been collaborating to monitor and protect the Common Swift colony occupying the town hall building, which has a swift-tower with 105 cells (Manica et al., 2019).

Monitoring was run in parallel with a ringing project, thanks to which 469 individuals have been ringed in recent years. Data on the reproductive phenology and reproductive success of the colony has been collected since the study began (2018).



Breeding phenology, reproductive success and survival of Common Swift (*Apus apus*) from a Northern Italian colony

Milo Manica[1-2], Daniela Casola[2], Stocchetti Alessandra[2], Manuela Bazzarelli[3], Davide Biganzoli[4], Michelangelo Morganti[5-6]

1 LIPU Birdlife Italia, via Udine 3/A – 43122 Parma, 2 Gruppo Insubrico di Ornitologia, via Manzoni 21 – 21050 Clivio (VA), 3 via Puccini 8 – 21023 Besozzo (VA), 4 Comune di Jerago con Orago, via indipendenza 17, 5 CNR-IRSA Via del Mulino 19, 20861 Brugherio (MB), 6 GreenRes Sezione di Zoologia del MUSE, Corso del Lavoro e della Scienza, 3 – 38122, Trento (TN)

Introduction

Matherials and methods

Results

Conclusion

Material and methods

- Weekly colony surveys were carried out over the 2018-2021 breeding seasons by checking each of the 105 cells.
- Number of eggs and chicks were registered.
- A total of 148 adults and 323 chicks were ringed over the four seasons (**Fig. 1**).
- Data analysis of reproductive success was carried out using Python.
- CMR (capture-mark-recapture) analysis of the was carried out using MARK v9.0.



Fig. 1. Ringing a breeding adult, on the terrace at the top of the swift nesting tower (photo: Milo Manica).

Breeding phenology, reproductive success and survival of Common Swift (*Apus apus*) from a Northern Italian colony

Milo Manica[1-2], Daniela Casola[2], Stocchetti Alessandra[2], Manuela Bazzarelli[3], Davide Biganzoli[4], Michelangelo Morganti[5-6]

1 LIPU Birdlife Italia, via Udine 3/A – 43122 Parma, 2 Gruppo Insubrico di Ornitologia, via Manzoni 21 – 21050 Clivio (VA), 3 via Puccini 8 – 21023 Besozzo (VA), 4 Comune di Jerago con Orago, via indipendenza 17, 5 CNR-IRSA Via del Mulino 19, 20861 Brugherio (MB), 6 GreenRes Sezione di Zoologia del MUSE, Corso del Lavoro e della Scienza, 3 – 38122, Trento (TN)

Introduction

Matherials and methods

Results

Conclusion

Results

- Colony size: 31 breeding pairs in 2018, 32 in 2019, 36 in 2020 and 38 in 2021.
- Phenology: we calculated First Arrival Day and Median Arrival Date. FAD: 12 April in 2018, 9 April in 2019 and 19 April in 2021. Due to the national lockdown, no data were recorded in April 2020. MAD: 31 May in 2018, 8 June in 2019, 29 May in 2020 and 2 June in 2021. The number of eggs laid peaked in the third ten day period in May while the number of chicks present peaked in the third ten day period in June (**Fig. 2**).

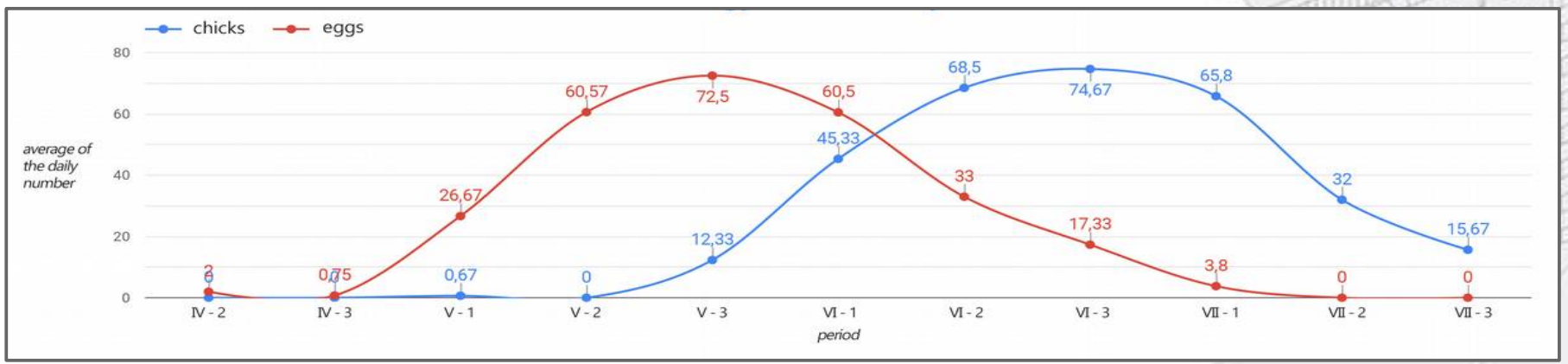


Fig. 2. The reproductive phenology of the colony obtained with the interpolation of the mean values over four years.

Breeding phenology, reproductive success and survival of Common Swift (*Apus apus*) from a Northern Italian colony

Milo Manica[1-2], Daniela Casola[2], Stocchetti Alessandra[2], Manuela Bazzarelli[3], Davide Biganzoli[4], Michelangelo Morganti[5-6]

1 LIPU Birdlife Italia, via Udine 3/A – 43122 Parma, 2 Gruppo Insubrico di Ornitologia, via Manzoni 21 – 21050 Clivio (VA), 3 via Puccini 8 – 21023 Besozzo (VA), 4 Comune di Jerago con Orago, via indipendenza 17, 5 CNR-IRSA Via del Mulino 19, 20861 Brugherio (MB), 6 GreenRes Sezione di Zoologia del MUSE, Corso del Lavoro e della Scienza, 3 – 38122, Trento (TN)

Introduction

Matherials and methods

Results

Conclusion

• Reproductive success was estimated as the proportion of the total number of fledged chicks over the total number of eggs laid. In 2018 it was: 61.47%, in 2019: 59.48%, in 2020: 54.97%, in 2021: 61.74%.

• Nest site fidelity (**Tab. 1**):

Year by year loyalty for each cell: Mean nesting cell fidelity was of 76.17% (±7.4) and was significantly different among years (p-value= 0.037). If we consider 3 consecutive years, the average fidelity is slightly reduced to 65%.

Year by year loyalty of a breeding pair to its cell and partner: the average percentage of pairs who return the following year with the same partner is 6.6%, a very low value when compared to that of swifts that return the following year with another partner (average of 17%). Returning couples seem to be faithful to their own cell. However, the result is not statistically significant.

• True survival (accounting for recapture probability) was as constant over the four years and was equal to 71.6% (±7.29). Recapture probability varied over the years: 2018 to 2019: ϕ= 0.458 (±0.096); 2019 to 2020: ϕ=0.331 (±0.074); 2020 to 2021: ϕ=0.722 (±0.153).

Swift fidelity to its own cell (n= 58)

	2019	2020	2021
Retour (%)	32.65%	20.21%	35.34%
Fidelity (%)	68.75%	73.68%	56.09%
	(compared to 2018)	*(compared to 2018-19)*	*(compared to 2018-20)*

Swift's fidelity to its partner / Pair's fidelity to its cell

	2019	2020	2021
Returning couples loyal to their mate (%)	5.80%	0.00%	7.10%
Returning couples loyal to their cell (%)	100.00% *	\	0.00%
	(compared to 2018)	*(compared to 2018-19)*	*(compared to 2018-20)*

* data not sufficiently significant

Tab. 1. Swift fidelity to its own cell and to its partner.

Breeding phenology, reproductive success and survival of Common Swift (*Apus apus*) from a Northern Italian colony

Milo Manica[1-2], Daniela Casola[2], Stocchetti Alessandra[2], Manuela Bazzarelli[3], Davide Biganzoli[4], Michelangelo Morganti[5-6]

1 LIPU Birdlife Italia, via Udine 3/A – 43122 Parma, 2 Gruppo Insubrico di Ornitologia, via Manzoni 21 – 21050 Clivio (VA), 3 via Puccini 8 – 21023 Besozzo (VA), 4 Comune di Jerago con Orago, via indipendenza 17, 5 CNR-IRSA Via del Mulino 19, 20861 Brugherio (MB), 6 GreenRes Sezione di Zoologia del MUSE, Corso del Lavoro e della Scienza, 3 – 38122, Trento (TN)

Introduction

Matherials and methods

Results

Conclusion

Discussion and Conclusions

The project showed encouraging results. A good picture of the reproductive phenology of the colony was obtained.
The data of reproductive success was close to that expected while the survival probability was very high when compared to other studies (e.g. Morganti et al., 2018). We did not observe any effects of the lockdown period, which differed from other studies and systems that considered the swifts (Manenti et al., 2020). The continuation of the ringing project in the following years will allow these and other parameters to be determined more accurately and to relate them to the reproductive success of previous years.

Acknowledgements

Thanks are due to LIPU for the figures from Costante Cavallaro and Stefania Villa; to GIO and GreenRes and their board of directors; to the Municipality of Jerago con Orago and the mayor Emilio Aliverti; to Alison Parnell for revising the language; and to all the people who helped with data collection; to Giacomo Baranzini for the graphic.

Bibliography

Manenti R., Mori E., Di Canio V., Mercurio S., Picone M., Caffi M., Brambilla M., Ficetola G.F., Rubolini D., 2020. *The good, the bad and the ugly of COVID-19 lockdown effects on wildlife conservation: Insights from the first European locked down country*. Biological Conservation

Manica M., Morganti M., Casola D., Bazzarelli M., Bogliani G., 2019. *Studio preliminare della fenologia e del successo riproduttivo in una colonia di rondone comune* (Apus apus) *a Jerago con Orago (VA).* Atti XX CIO, Napoli

Morganti M., Rubolini D., Åkesson S., Bermejo A., de la Puente J., Lardelli R., Liechti F., Boano G., Tomassetto E., Ferri M., Caffi M., Saino N., Ambrosini R., 2018. *Effect of light-level geolocators on apparent survival of two highly aerial swift species*. Journal of Avian Biology